HYDRA Documentation

HYDRA MW4.0pe

Preparation Guide for Windows Server 2019

Version 1.0.23049

Last changed on: 02.09.2020

Copyright

Contents

1 Introduction

This configuration manual explains how to configure Windows Server 2019 for use as a server for MPDVs Manufacturing Integration Platform (**MIP**) based products like MIP 1.1, HYDRA MW4.0pe or FEDRA 1.1.

The configuration of the server should always be based on MPDVs hardware and software recommendations for the respective MIP based product.
See manuals: **HW_SW_GUIDE.pdf**

!ATTENTION!
The Server must be a dedicated server for running any MIP based products.
There must be no additional functionality used, e.g. like File Server, Print Server, OPC Server, Domain Controller (PDC, BDC), Active Directory Controller or similar.

If the configuration of your server deviates from the configuration described in this manual, additional efforts and expenses might be necessary for the installation of MIP based products.
Already agreed shipping or installation appointments might be delayed or may have to be postponed.

MIP based products might not be able to work properly on such a server.
It might even be impossible to perform a successful installation on such a server.

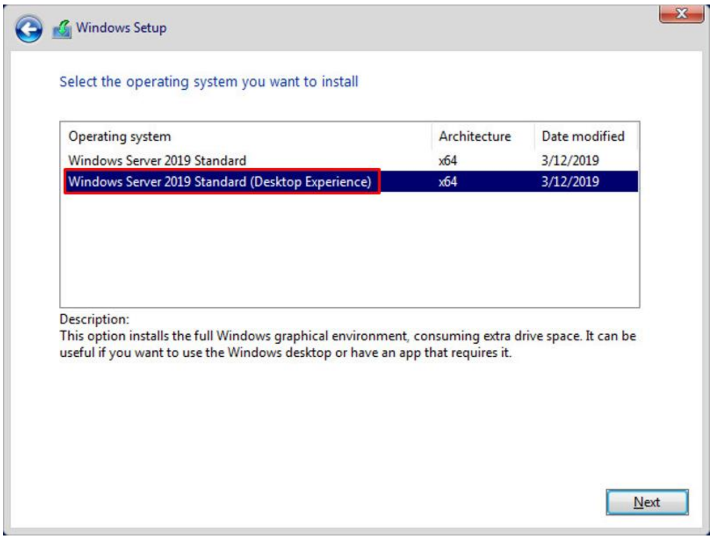
It might not be possible to install different MIP based products on the same server.

2 Operating System

- Server platform according to the hardware and software recommendations of MPDV Mikrolab GmbH is required.

- The Windows server must be a **dedicated server** for running any MIP based product.
  There must be no additional functionality be used, e.g. like Domain Controller (PDC, BDC), Active Directory Controller, File Server, Print Server, OPC Server or similar.
  In such cases it might be impossible to perform a successful installation of MIP based products.
  Pre installation tasks at a MPDV site might also not be possible or only with additional effort and expenses.

- Windows Server 2019 with a graphic user interface (**GUI**) is required.
  During the installation of Windows Server 2019 select the option "**Desktop Experience**":



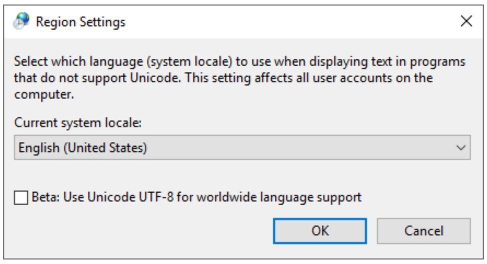
- The Windows desktop resolution should be set to **1920 x 1080** at least to **1280 x 1024**.

- The recent Windows Service Pack approved by MPDV Mikrolab GmbH for the respective MIP based product must be installed on the server. See manual for hardware and software recommendations.

- For information about hard disk requirements and partitioning see chapter “3 Hard Disk Layout”.
  All partitions must be formatted with file system **NTFS**.

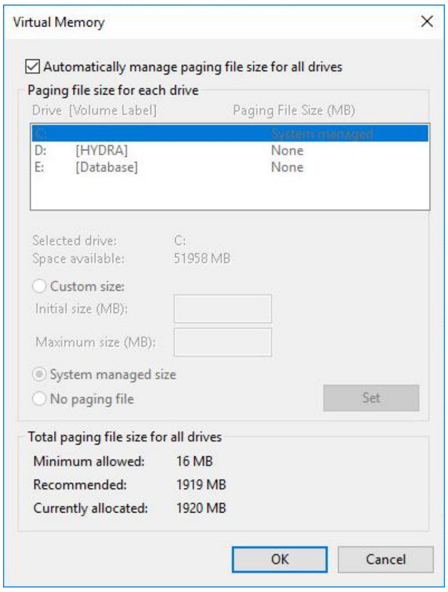
- Windows must be installed either with Desktop language **English** or **German**.
  The language versions of the operating system and the database must match!

- The system locale of Windows must either be set to
  "**English (United States)**" or "**German (Germany)**":

  If available, do **not** activate the option "Use Unicode UTF-8 for worldwide language support"!

- **Virtual Memory:**
  Automatically manage paging file size for all drives (= Windows Default, recommended by MPDV)

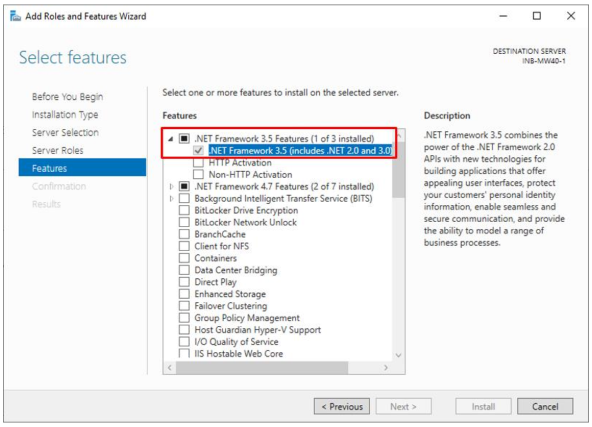
  

  If you choose to use individual settings, please make sure to follow the recommendations of the operating systems manufacturer to ensure a stable and performant system.

- **Network**
  TCP/IP network protocol (**IPv4**) based on an Ethernet network must be available.
  At least the following TCP/IP ports and port ranges must be available for MIP based products:
  1434, 1521, 3300-3399, 3299, 8080, 9000-9005, 10000, 10100, 10103, 10111, 10120-10127, 10150, 10177, 18080, 30101-30108
  If these ports (especially port 10000 for HYDRA) are not available there might be additional efforts necessary during the installation.
  Additional ports will be required when the MIP based product is supposed to be installed as a multi system installation (e.g. HYDRA) or when additional functionality should be used.

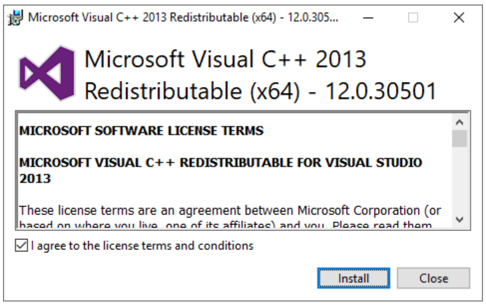
- Choose any IP address and hostname for the server which suits your needs.
  It is necessary that the MIP based product server uses a **dedicated IP address**.

- Installed Feature **.NET Framework 3.5 (includes .NET 2.0 and 3.0)** is required.
  Start-Button – Server Manager – Manage – Add Roles and Features

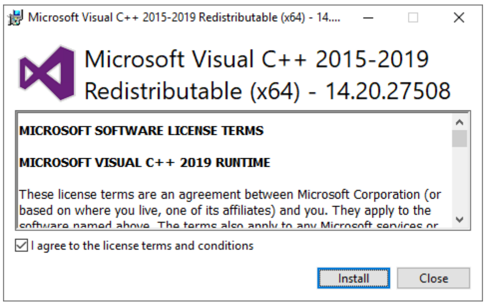


  To add this feature the directory **x:\sources\sxs** from the Windows installation DVD is needed.
  To avoid unnecessary delays during the installation please make sure this feature is activated prior to the installation date!

- Installed **Visual C++ 2013 Redistributable (x64)** minimum version **12.0.30501** is required, e.g.:
  https://www.microsoft.com/en-US/download/details.aspx?id=40784

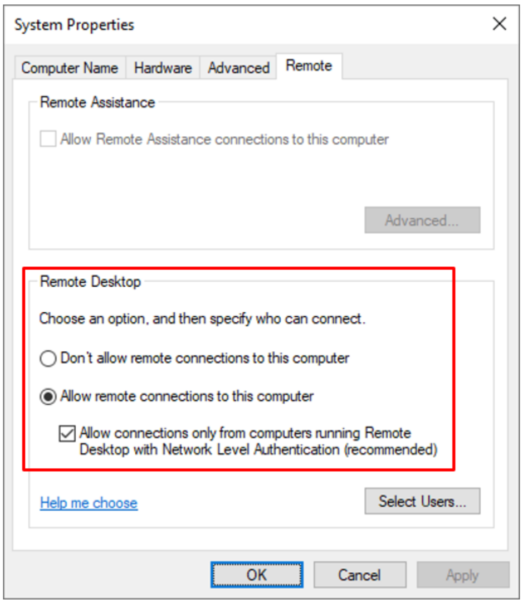


- Installed **Visual C++ 2015-2019 Redistributable (x64)** minimum version **14.20.27508** is required, e.g.:
  https://support.microsoft.com/en-us/help/2977003/the-latest-supported-visual-c-downloads



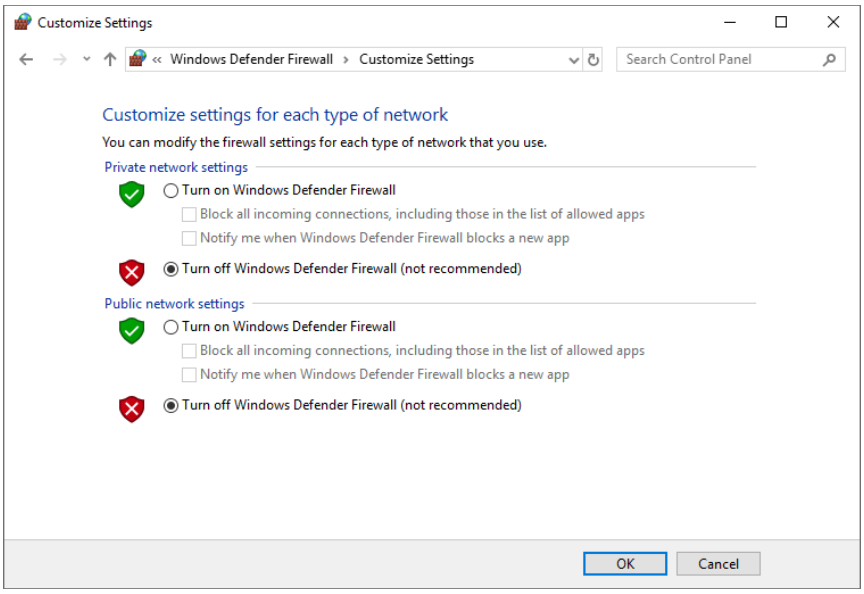
- Allow remote connections to the server:

  Start-Button – Control Panel – System and Security – System – Advanced system settings – Remote

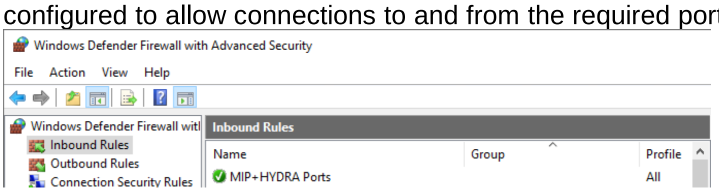


- The local Windows Firewall should be turned off:

  Start-Button – Control Panel – System and Security – Windows Defender Firewall – Turn Windows Defender Firewall on or off



If the local firewall needs to stay turned on there must be appropriate **Inbound** and **Outbound** rules configured to allow connections to and from the required ports of the MIP based product (see above):

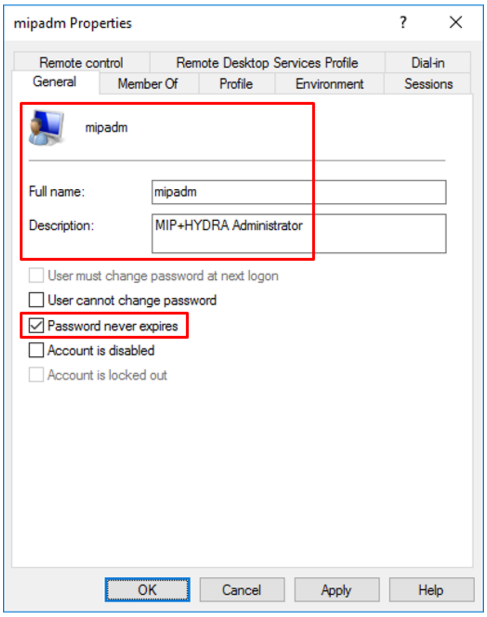


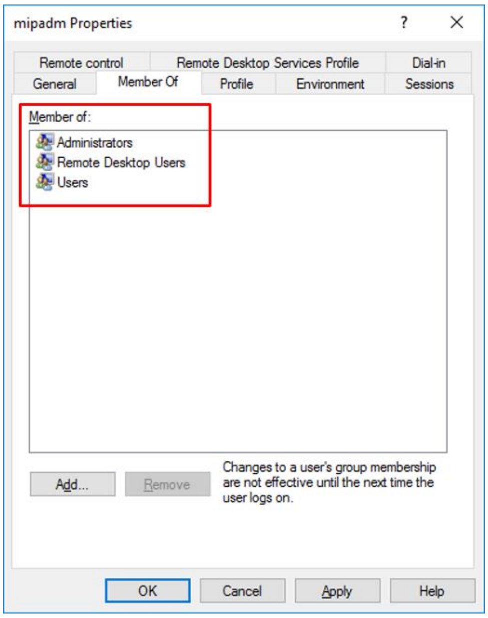
- **User mipadm**

  There should be a local user account "**mipadm**" with full name "**mipadm**" and an appropriate description, e.g. “**MIP Administrator**” or “**MIP+HYDRA Administrator**”.

  Alternatively to a local user account it would be possible to provide a domain user account "**mipadm**" from the customers Active Directory Domain Services.

  User "mipadm" must be at least a member of the local groups "**Administrators**", "**Remote Desktop Users**" and "**Users**".





The password for user “mipadm” must never expire and should be set to "**Mip74821**".

Optionally any other password can be used which must then be disclosed to the MPDV employees who are installing the MIP based product for you.

The following characters are **not** allowed for the password:

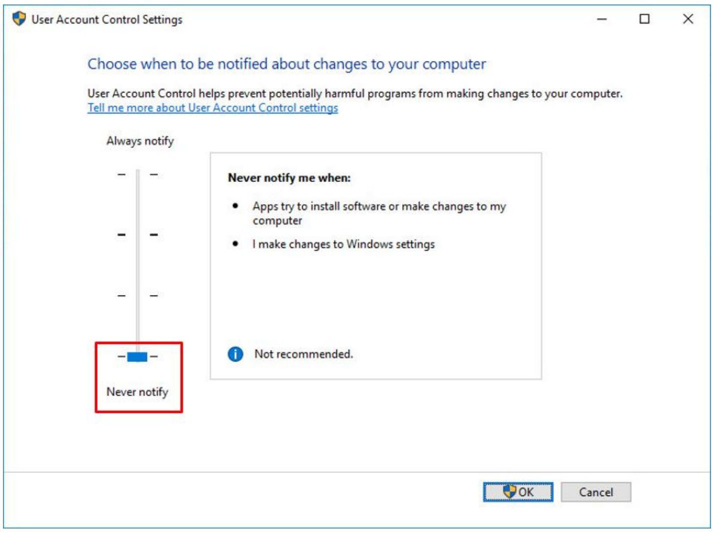
- Characters with ASCII Code **> 126** (e.g. German umlauts, French “umlauts”, etc.)

- Pipe**: |**

Periodic changes of the user’s password would be possible as long as you take into account that there will be at least one MIP service running on the server using the login credentials for user “mipadm”.

- User Account Control (UAC) **<u>must</u>** be deactivated.

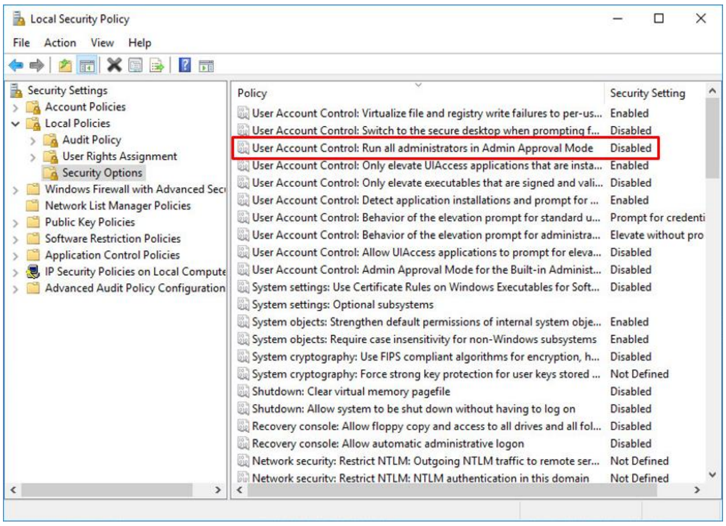
  Start-Button – Control Panel – User Accounts – User Accounts – Change User Account Control settings

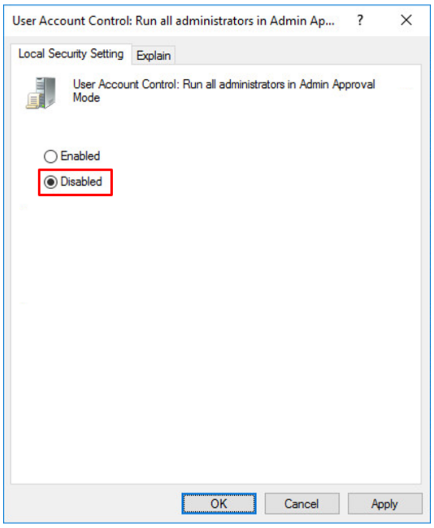


Never notify

- In “Local Security Policy” the following policy **<u>must</u>** be **disabled**:

  Start-Button – Windows Administrative Tools – Local Security Policy – Local Policies – Security Options – User Account Control: Run all administrators in Admin Approval Mode



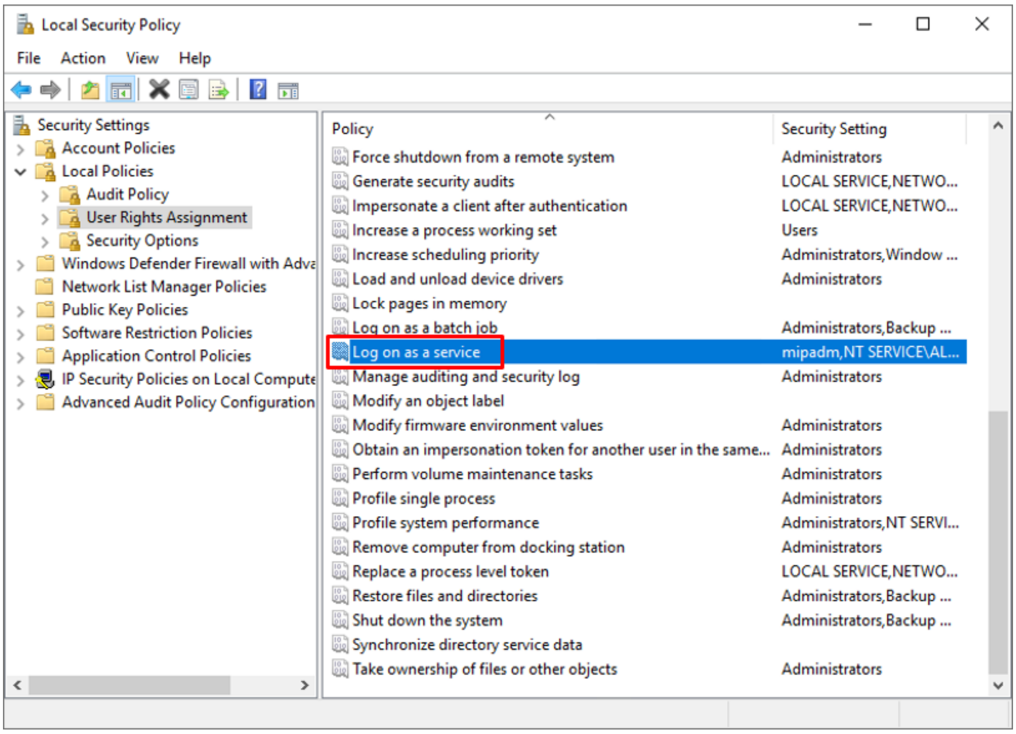


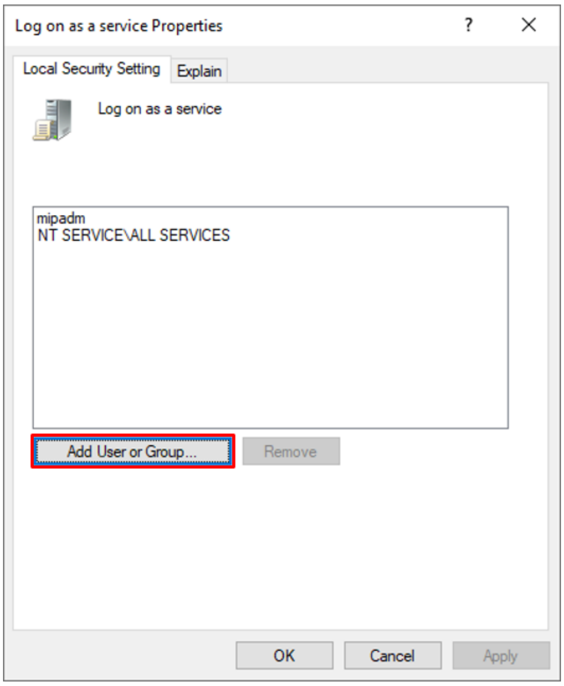
Disabled

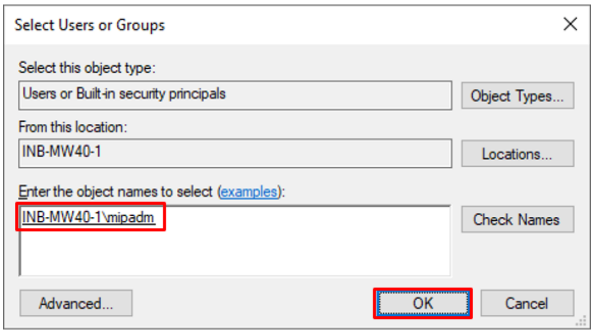
- **Log on as service**

  In “Local Security Policy” add the user “**mipadm**” to the “Log on as service Properties”

  Start-Button – Control Panel – System and Security – Administrative Tools – Local Security Policy – Local Policies – User Rights Assignment – Log on as a service

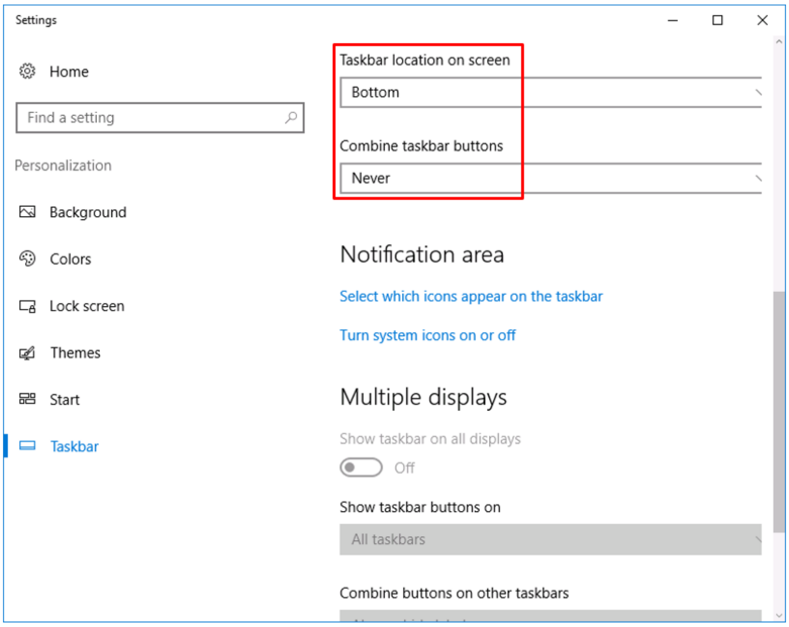






- Logon to the MIP server as local user "**mipadm**".

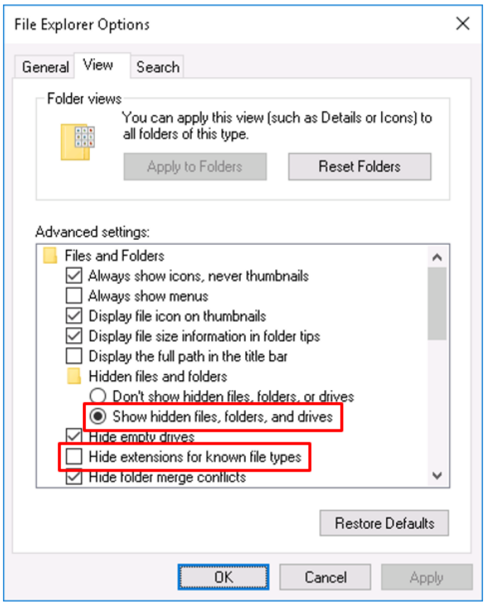
- **Start-Button – Control Panel – Appearance and Personalization – Taskbar and Navigation**



Taskbar location on screen: Bottom

Combine taskbar buttons: Never

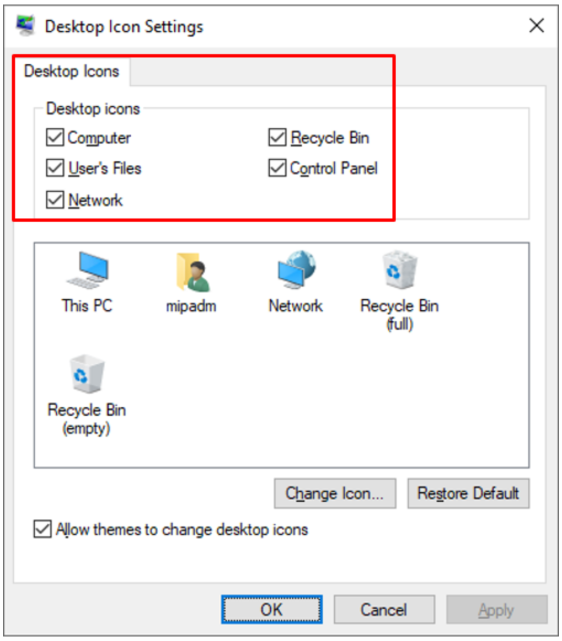
- **Start-Button – Control Panel – Appearance and Personalization – File Explorer Options – View**



Activate: Show hidden files, folders and drives

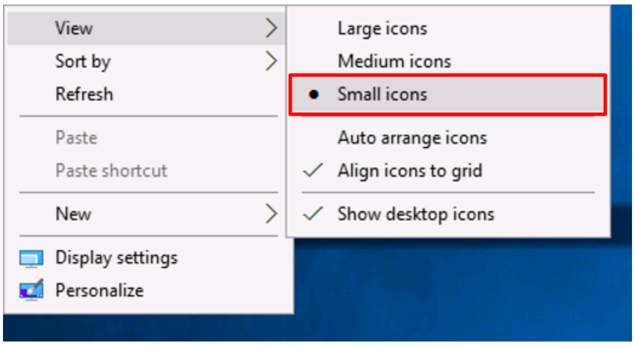
Deactivate: Hide extensions for known file types

- **Start-Button – Settings – Personalization – Themes – Desktop icon settings**



Activate all Desktop icons

- Right click on the Windows desktop and then in "**View**" select: "**Small icons**"



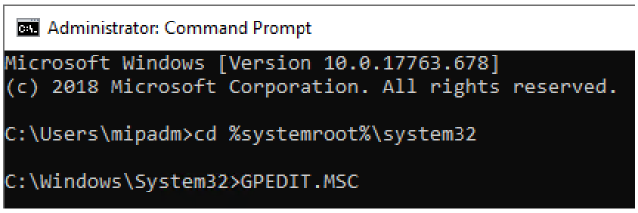
- Allow multiple Remote Desktop Sessions per User (**optionally if required**)

  Start-Button – Windows System – Command Prompt

  type in the following commands:

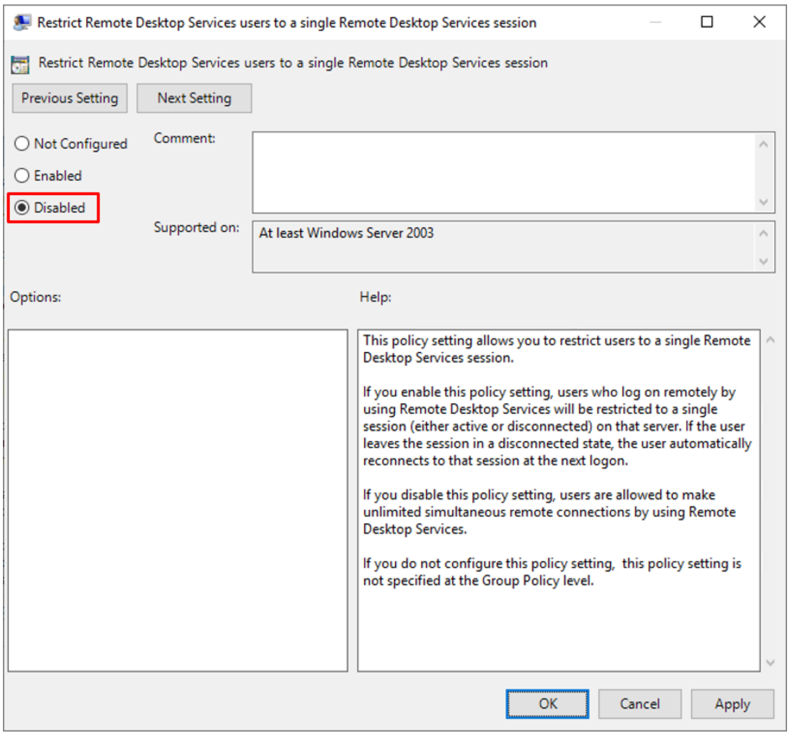
  `cd %systemroot%\system32`

  `GPEDIT.MSC`



In the Local Group Policy Editor make the following settings:

Local Computer Policy – Computer Configuration – Administrative Templates – Windows Components – Remote Desktop Services – Remote Desktop Session Host – Connections – Restrict Remote Desktop Services users to a single Remote Desktop Services session



Disabled

2.1 Web Browser Software

If you plan to use MPDV's product Smart MES Applications (SMA) version **8.2** you must be aware that Microsoft's Internet Explorer (IE) is no longer supported by MPDV.
You need to provide an alternative web browser software like **Google Chrome** which is MPDV's recommended web browser software for using SMA 8.2.
Please make sure that Google Chrome is installed on your Windows server.

3 Hard Disk Layout

The storage demand of a MIP based system and its database usually increases over time.
Therefore the available disk storage should be large enough right from the start.

For a MIP based system we recommend the following hard disk capacity to be available:
200 GB for the MIP based application including possible archive data to be generated in the future
400 GB for the MIP based product database (default configuration)

Apart from a sufficiently large Windows boot and system partition (usually **C:**) of at least **80-100 GB** there should be at least another partition **D:** available which must provide the above mentioned disk capacity of at least **600 GB**.

If there are additional hard disks and partitions available, e.g. like **E:\, F:\, G:\, H:** etc., the data files for the MIP based product database could be spread for performance reasons.
When using online backup solutions for the database software the use of a dedicated hard disk and partition is highly recommended.

All partitions must be formatted with file system **NTFS**.

CD/DVD drives should use a drive letter at the end of the available range, e.g.: **Z:**.

Configuration example for a Windows server with 5 hard disks (each 500 GB in size):

1. hard disk:

C:\ NTFS	100 GB	Windows, tools and pagefile.sys
D:\ NTFS	400 GB	MIP based product and database application

2. hard disk:

E:\ NTFS	500 GB	database data files

3. hard disk:

F:\ NTFS	500 GB	database data files

4. hard disk:

G:\ NTFS	500 GB	database data files

5. hard disk (additional disk for online backup solutions):

H:\ NTFS	500 GB	backup files for online backup solutions

4 Database ORACLE

For information about currently supported database versions please consult the recent recommendations regarding hardware and software by MPDV Mikrolab GmbH.

If the Oracle database software is supposed to be preinstalled by the customer himself, please contact MPDV Mikrolab GmbH first.
You will then be provided with the proper database installation and configuration manual(s) which will allow you to do the installation and configuration according to the needs of the MIP based product.

If you plan to have separate application and database servers the 64Bit Oracle client software needs to be installed on the MIP application server.

All MIP based product databases are supposed to use the character set **AL32UTF8**.
The use of other character sets is not supported and may cause problems with the MIP based application.

Should the MIP based product be installed on a Windows server already running one or more Oracle databases, it is necessary that NLS_LANG in the Windows registry matches the requirements for the MIP based product:
Key: **NLS_LANG=AMERICAN_AMERICA.AL32UTF8**
If that is not possible, there is no way to install the MIP based product on such a server for the time being.

The Archive Log Mode for Oracle databases installed by MPDV Mikrolab GmbH is **deactivated** by default. This will prevent an unnoticed overflow of file systems and therefore the halt of all database functionality. The Archive Log Mode will be activated by MPDV Mikrolab GmbH on special request by the customer only.

When using online backup solutions with an activated Archive Log Mode the use of a dedicated hard disk is highly recommended.
See chapter "3 Hard Disk Layout".

5 Database Microsoft SQL Server

For information about currently supported database versions please consult the recent recommendations regarding hardware and software by MPDV Mikrolab GmbH.

If the SQL Server software is supposed to be preinstalled by the customer himself, please contact MPDV Mikrolab GmbH first.
You will then be provided with the proper database installation and configuration manual(s) which will allow you to do the installation and configuration according to the needs of the MIP based product.

If you plan to have separate application and database servers the SQL Server Native Client software and the SQL Server Management Studio need to be installed on the MIP application server.

The language versions of the operating system and the database must match (e.g.. English – English or German – German).
Mixed language versions are not allowed and supported by Microsoft.

For performance reasons the write cache of the hard disk controller or the hard disks should be activated.

The recovery model for SQL Server databases installed by MPDV Mikrolab GmbH is initially set to "**Simple**" by default.
This will prevent an unnoticed overflow of the transaction log file and subsequently of file systems and therefore the halt of all database functionality.
Recovery model “Full” will be activated by MPDV Mikrolab GmbH on special request by the customer only.

When using online backup solutions with recovery model “Full”, the use of a dedicated hard disk is highly recommended.
See chapter “3 Hard Disk Layout”.